

# Alejandro Sánchez Yalí

## Product Engineer



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## SUMMARY

Product Engineer who takes products from an idea to a deployed system alone. In the last two years built three complete products – Plixiq (AI customer support on WhatsApp), Aluna (AI-assisted recruitment) and VitaStock (surgical supply chain for private clinics) – owning discovery with users, architecture, backend, frontend and deploy. 8+ years of engineering, an M.Sc. in Mathematics and 11 years teaching, which is why the hardest parts of the job – explaining a system and deciding what not to build – come naturally.

## TECHNICAL SKILLS

- **Ship the whole thing:** Python, FastAPI, Django, TypeScript, Next.js, React, Node.js, NestJS, PostgreSQL, Redis, Docker, Railway, AWS
- **AI in products:** LLMs, LangChain, LangGraph, LiteLLM, RAG, MCP, OpenAI API, Whisper, ElevenLabs, WhatsApp Cloud API
- **Decide well:** Clean architecture, DDD, SOLID, modular monolith, event-driven design, trade-off analysis, technical design sessions
- **Keep it alive:** GitHub Actions, CI/CD, Docker, background jobs, real-time dashboards, TDD, Vitest, Jest, code review

## PRODUCTS SHIPPED

- **Plixiq – AI Customer Support on WhatsApp** – *Python, FastAPI, LiteLLM, PostgreSQL, Next.js, Railway*
  - Took the product from an idea to a deployed system alone: discovery, architecture, backend, frontend and deploy
  - About 40,000 lines to a near-complete MVP in 4 months part-time, running at \$60–80/month because cost scales with LLM usage rather than infrastructure
- **Aluna – AI Recruitment Platform for Staffing Agencies** – *Next.js 15, FastAPI, LiteLLM, PostgreSQL, Inngest, Claude, Gemini, WhatsApp Cloud API*
  - Lead engineer of an AI recruitment platform for staffing agencies, built in 3 months: CVs are scored automatically, an agent interviews candidates over WhatsApp or web chat, and the recruiter gets a shortlist and a client-ready report
  - Deliberate model routing through LiteLLM (Gemini to read documents, Claude to converse), multi-tenant roles from a single policy source, and Ley 1581/2012 consent, deletion receipts and audit logs built into the data model
- **VitaStock – Surgical Supply Chain for Clinics** – *Python, FastAPI, Next.js, TypeScript, PostgreSQL, Docker, Railway*
  - Sole architect and engineer of a platform commissioned by a private surgical clinic in Colombia – 12 modules and 5 user roles designed to replace spreadsheet-based inventory and procurement
  - Surgery scheduling with supply baskets and real-time consumption tracking, lot-level traceability with expiry alerts, and true per-patient cost through a cardex

## EXPERIENCE

**Independent Software & AI Consultant**  
*Freelance*

Aug 2024 – Present  
*Remote*

- Built and deployed three complete products in two years, each owned end to end – from the first conversation with the customer through to the deploy
- Led technical design sessions with clients, turning vague business problems into architectures and naming the trade-off behind every technology choice
- Use Claude Code daily on live client projects, which raised throughput enough that review and taste became the constraint rather than typing
- Code reviewer for a client’s development team and architect of their workflow automation platform’s database design

## AI Specialist

*Lapzo*

Aug 2025 – Present

*Remote, Mexico*

- Designed and built a Digital Professor that converses with learners in real time and teaches through interactive slides
- Made the key architectural calls: which LLM orchestration stack, how voice synthesis plugs into the conversation, where the service boundaries go

## Full Stack Engineer

*Monadical*

Apr 2021 – Aug 2024

*Remote, Canada/US*

- Three years on a distributed team across Canada, the US and Latin America, shipping full stack products in Python and TypeScript
- Participated in system design and architectural decisions, weighing performance, scalability and maintainability
- Led frontend architecture and testing strategy; reviewed code and mentored engineers

## Full Stack Engineer

*BCFort*

Aug 2018 – Oct 2020

*Medellín, Colombia*

- Designed system architecture for blockchain and analytics platforms, from smart contract patterns to service boundaries
- Built decentralised applications and NFT marketplaces with React, Web3.js, Ethereum and Hyperledger

## Professor of Mathematics & Machine Learning

*Universidad de Antioquia*

Jan 2010 – Oct 2021

*Medellín, Colombia*

- Eleven years teaching Machine Learning, Data Science and Mathematics, and building the AI/ML elective curriculum for the Mathematics Institute
- Mentored student research projects – the same work that later became code review and design docs

## OPEN SOURCE

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- [talking-avatar-with-ai](#)  – 464 stars, 110 forks. Open-source conversational AI avatar: OpenAI GPT, Whisper, ElevenLabs, Rhubarb lip-sync, Three.js
- [Monadical-SAS/Morpheus](#)  – 19 pull requests. Open-source AI image generation platform: FastAPI, React, Stable Diffusion
- 493 GitHub stars across own open-source projects – top 4.8% of ranked GitHub profiles (GitRanks)

## TALKS & WRITING

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- [Scaling Learning Models: Parallelism Strategies in JAX with Flax](#)  – PyCon Colombia, 2024
- [Building Plixiq: a Multi-Tenant WhatsApp AI Support Platform](#) – architecture case study (2026)
- [Revolutionizing Animation: Building Digital Humans with LLMs](#) (2024)

## EDUCATION

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### M.Sc. in Mathematics

*Universidad de Antioquia*

Aug 2010 - Sep 2013

*Medellín, Colombia*

### B.Sc. in Teaching Mathematics and Physics

*Universidad de Antioquia*

Jan 2004 - Oct 2009

*Medellín, Colombia*